File Systems and Partitioning

Objectives:

At the end of this episode, I will be able to:

1. Identify the appropriate file system for a given server installation. Compare and contrast file systems and partition types.

Exam Objective: *2.1 - Given a scenario, install server operating systems.*

Description: In this episode, the viewer will identify a variety of file systems, and partitioning schemes, the associated considerations as well as determining which to choose for a given server installation.

- Partitions and volumes
    - Defined by boot scheme
- File system types
    - ext4
        - Journaling file system developed for Linux systems, succeeding ext3
        - Fully released in Linux Kernel 2.6.28
        - 1 EB volume size
        - 16 TB file size
        - Support for POSIX permissions and POSIX ACLs
        - Support for encryption
    - New technology file system (NTFS)
        - Journaling file system developed for Windows-based systems
        - 256 TB volume size (Windows Server 2016 or earlier)
        - 8 PB volume size (Windows Server 2019 or later)
        - Support for permissions and ACLs
        - Supports compression and encryption
    - Resilient file system (ReFS)
    - VMware file system (VMFS)
        - Clustered file system
        - Designed for VM image storage
        - 64 TB volume sizes
        - 62 TB file sizes
        - Current version: VMFS6
        - Support
            - VMFS5/VMFS6 can be seen by ESXi 6.5 hosts and later
            - VMFS5 can be seen by ESXi 6.0 host and earlier
    - Z file system (ZFS)
        - Developed by Sun Microsystems (aquired by Oracle)
        - Designed for long term storage (implements mechanisms to check for and correct errors
        - Supports rollback of recent changes
        - 128-bit file system
        - 256 quadrillion ZB volume sizes
        - 16 EB file size

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- ext2,3,4 - https://opensource.com/article/17/5/introduction-ext4-filesystem
- NTFS - https://docs.microsoft.com/en-us/windows-server/storage/file-server/ntfs-overview
- ReFS - https://docs.microsoft.com/en-us/windows-server/storage/refs/refs-overview
- VMFS - https://docs.vmware.com/en/VMware-vSphere/7.0/com.vmware.vsphere.storage.doc/GUID-5EE84941-366D-4D37-8B7B-767D08928888.html
- ZFS - https://docs.oracle.com/cd/E19253-01/819-5461/zfsover-2/